

Step 2: Features, Personas, Scenarios and User Stories

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2.1 Purpose of Step 2

The purpose of Step 2 is to turn the product vision into clear and practical requirements. Based on the vision of an integrated services app for SETU Computing and Mathematics students, this step defines:

The main product features
typical users (personas)
real usage scenarios
user stories

The goal is to ensure the system is designed to be useful for all stakeholders:

- students (real experience and exam success)
- the university (better results and reputation)
- Small businesses (automation and practical solutions).

2.2 How the Feature List Was Derived

The feature list was derived from the two main needs identified by the team:

1. **Exam performance needs:** students want to achieve the highest possible exam results using a preparation system that is built directly on past exam questions and marking schemes.
2. **Assignment usefulness need:** students want assignments to be real and useful (not imaginary ideas), by completing projects to help small businesses and gain real client-style experience.

These needs are fully aligned with the product vision pillars described in Step 1

- educational support
- Business integration support
- social collaboration
- career preparation

2.3 Feature List

2.3.1 Educational Support (Exam Preparation Engine)

Core idea: Exam preparation is built around past exam questions. The system minimizes cognitive overload and focuses on understanding the method, not long calculations.

Key features:

- **Subject selection:** the student selects a module/subject for preparation.
- **Assessment structure view:** the student sees the breakdown (e.g., Exam 50%, Assignment 50%).
- **Exam preparation mode** based on past papers:
 - Questions are **sorted by frequency** across previous years.
 - Questions are grouped by **difficulty**.
- **Two-level preparation strategy:**
 - Level 1 (Fast Pass Strategy):**
Focus on questions that appeared in more than 75% of past exam papers of years, to prepare quickly for a pass.
 - Level 2 (High Grade Strategy):**
includes more difficult topics, longer videos, and less frequent questions for a higher grade.
- **Micro-learning videos:**
 - For high-frequency questions: very short videos (around 1 minute) with a simple explanation of a similar example.
 - For harder topics: longer videos with deeper explanation.
- **Minimal mental load design:**

- Where possible, questions are delivered as **simple tests (MCQ)** instead of complex input.
- After each video, students get **very simplified examples** that show the method clearly and avoid unnecessary calculations.
- **Visual simulations** (where relevant):
 - For example, vector addition can be shown in a simulator with the key formula visible.
- **Score-based tracking using marking schemes:**
 - The system uses the marks from past exam papers/marking schemes to calculate progress.
- **Progress analytics (core outputs):**
 - Exam coverage percentage (how much of the exam the student has covered).
 - Estimated probability of passing (e.g., after finishing Level 1 the system can show an >75% pass probability based on past-year patterns).
 - Recommendation of the next best step (what to study next).

2.3.2 Business Integration Support (Assignments with Small Businesses)

Core idea: assignments should be useful and real. Instead of inventing a fake topic (e.g., “create a website for an imaginary business”), students choose a real small business topic from a list and build a real solution as their assignment.

Key features:

Small business project list: curated list of small businesses that need practical digital solutions, such as:

- Website development
- Online ordering systems
- Booking and appointment systems
- Simple databases for customer or inventory management
- Basic CRM systems
- Marketing support (social media integration, email campaigns, promotional pages)
- Automation of internal processes
- Data organisation and reporting tools

These projects allow students to work on real business problems such as order management, customer tracking, digital presence, and business process automation.

Multiple student solutions per business: one business can receive several student projects.

Business chooses the preferred solution: after review, the business can select the project they like most to start further development.

- **Coordinator-based communication model:**
 - Students do not directly contact the business.
 - Dedicated coordinators with real experience communicate with the business, collect requirements/content, and provide structured information to students.
 - This prevents business users from being overloaded or disturbed in an unprofessional way.
- **Team building from the best work:**
 - Based on the strongest student projects, teams can be formed.
 - The system is flexible: teams can be created to finish one project, become longer-term teams, or join other business projects.
- **No change to academic grading rules:**
 - The university assessment process remains unchanged.
 - The platform only makes the assignment work more realistic and more beneficial.

2.4 Personas

Persona 1: Mark – Fast-Pass Exam Student

Mark, age 21, is a second-year Computing student at SETU. He lives in shared accommodation near campus and works part-time in a supermarket. Because of his work schedule, he often has limited time for structured study.

Mark is comfortable using digital tools and online platforms, but he prefers simple and clear interfaces. He does not like overly complex systems or long theoretical explanations. In previous semesters, he struggled with exam preparation because he focused on reading lecture slides rather than analysing past exam questions.

Mark's main goal is to pass his exams efficiently and reduce stress. He is not initially aiming for a very high grade; instead, he wants a reliable and structured way to secure the minimum required mark. He is especially interested in a preparation method that shows which questions appear most frequently and what is most likely to be examined.

He would use the platform primarily in "Level 1 – Fast Pass" mode to focus on high-frequency exam questions, short videos, simplified examples, and clear progress indicators showing his probability of passing.

Persona 2: Sarah – High-Grade Exam Student

Sarah, age 22, is a third-year Mathematics and Computing student at SETU. She is academically motivated and plans to apply for a postgraduate programme after graduation. She consistently achieves good results but wants to improve her performance further.

Sarah is confident with digital tools and frequently uses online learning platforms, coding environments, and mathematical software. She prefers structured learning materials and detailed explanations rather than surface-level summaries. She is willing to spend more time studying if she believes it will significantly improve her results.

In previous semesters, Sarah performed well in coursework but sometimes lost marks in exams due to unexpected or more complex questions. She wants a preparation system that goes beyond basic revision and includes deeper analysis of less frequent and more difficult topics.

Her goal is to achieve a high grade (above 85%) and to understand not only common exam questions but also advanced variations. She would primarily use “Level 2 – High Grade Strategy”, focusing on:

- harder and less frequent past exam questions,
- longer and deeper explanatory videos,
- detailed analytics showing coverage and predicted grade,
- clear recommendations on which advanced topics to revise next.

Sarah values accuracy, structured feedback, and measurable progress indicators that help her optimise her study time.

Persona 3: Daniel – Assignment-to-Portfolio Student

Daniel, age 23, is a final-year Software Development student at SETU. He is focused on employment after graduation and is actively preparing his CV and LinkedIn profile. He understands that employers value practical experience more than theoretical assignments.

Daniel is confident in programming and web development but has limited experience working with real clients. Most of his previous assignments were based on imaginary project ideas, which made it difficult to demonstrate real-world impact in his portfolio.

He wants his academic work to be meaningful and aligned with industry expectations. He is particularly interested in building real solutions that can be deployed and used by actual businesses.

Daniel’s main goal is to complete assignments that:

- solve real business problems,
- involve structured requirements,
- simulate real client communication,

- produce portfolio-ready results.

He would use the platform to:

- browse the small business project list,
- receive structured requirements from coordinators,
- collaborate with other strong students,
- submit a professional solution,
- potentially join a team formed from the best projects.

Daniel values realism, practical outcomes, and opportunities to demonstrate his skills to future employers.

Persona 4: Olivia – Small Business Partner (Indirect User)

Olivia, age 38, is the owner of a small beauty and wellness salon in Waterford. She manages daily operations, customer bookings, staff schedules, and marketing activities. Most of her time is focused on serving clients and managing appointments.

Olivia has basic digital skills. She uses social media platforms to promote her business and relies on simple tools such as email and spreadsheets. However, she does not have technical knowledge of web development or database systems. She cannot dedicate large amounts of time to complex technical discussions.

Olivia recognises that her business could benefit from improved digital tools, such as:

- a modern website,
- an online booking system,
- better customer data organisation,
- improved digital marketing support.

Her main goal is to receive a useful and practical digital solution without being overwhelmed by multiple student contacts or technical complexity.

She would use the platform indirectly through coordinators who:

- collect her business requirements,
- structure them clearly,
- manage communication professionally.

Olivia would log into the platform only to:

- review submitted student solutions,
- compare options in a simple dashboard,
- select the preferred version for further development.

She values professionalism, clarity, time efficiency, and minimal disruption to her daily business activities.

2.5 Scenarios

Scenario 1: Level 1 Fast Pass Exam Preparation

A student selects a subject and chooses “Exam”. The system shows Level 1 and explains that it is based on questions that appeared in 75% years. The student watches short videos, completes simple tests, and solves simplified examples. After completing Level 1, the system shows progress and an estimated pass probability based on past-year frequency patterns.

Scenario 2: Level 2 High Grade Preparation

A student completes Level 1 and then chooses Level 2 to target a higher grade. The student studies harder topics with longer videos and works through more difficult past-exam problems. The system continues to calculate coverage and recommends the best next topic.

Scenario3: Assignment Based on a Real Small Business

A student is required to complete a module assignment, such as:

- developing a website for a Web Development module,
- creating a database for a Database Systems module,
- Implementing a small software solution for a Programming module.

Instead of inventing an imaginary project idea, the student selects a real small business from the platform’s project list.

The coordinators communicate with the business and collect all necessary information, such as:

- business description,
- required features,
- content (text, images, services),
- technical limitations or preferences.

This structured information is then provided to the students through the platform.

Multiple students may work on solutions for the same business. Each student (or group) submits their own version of the project.

After reviewing the submitted solutions, the business selects the preferred version. The selected project may be further improved and deployed in practice.

Based on the strongest student submissions, teams can be formed to continue development with mentors, refine the solution, or participate in future business projects. The system remains flexible: teams may work on a single project or continue collaborating long-term.

This approach ensures that:

- students gain real experience working on practical problems,
- The university improves the quality and relevance of assignments,
- Small businesses receive useful digital solutions without being overloaded by direct student communication.

2.6 User Stories

Exam Preparation Stories

- As a student, I want to choose a subject and see the exam/assignment breakdown so that I understand what I need to focus on.
- As a student, I want Level 1 preparation based on questions that appeared in >75% years so that I can prepare quickly for a pass.
- As a student, I want short videos and simplified examples so that I can learn the method without cognitive overload.
- As a student, I want simple test-style questions where possible so that I can practise without complex input.
- As a student, I want my progress to be calculated using real marking scheme marks so that my preparation is realistic.
- As a student, I want to see exam coverage, pass probability, and the next recommended topic so that I know what to do next.

Assignment + Business Stories

-As a student, I want a list of small businesses needing websites so that my assignment work is useful and realistic.

-As a student, I want coordinators to handle business communication so that the business is not overloaded and the requirements are clear.

-As a business partner, I want to choose the best student solution so that I receive a result that matches my needs.

-As a coordinator, I want to collect requirements and provide structured content to students so that projects run smoothly.

-As a student team, we want to be formed based on the best work so that we can deliver higher-quality solutions and gain real experience.

2.7 Feature List Revisited (Summary)

After personas and scenarios were defined, the most important features were confirmed as:

- Frequency-based exam question sorting
- Level 1 and Level 2 preparation modes
- Micro-videos + simplified examples (low cognitive load)
- Marking-scheme-based scoring and analytics (coverage + pass probability + next-step recommendation)
- Business project list with coordinator-managed communication
- Business selection of preferred solution
- Flexible team creation from best work